

# Deep Inelastic Scattering Revisited

Felix Hekhorn  
University of Jyväskylä

Lecture notes for the Jyväskylä Summer School 2026



# Abstract

Discuss Yadism [\[1\]](#).



# Contents

|          |                                    |           |
|----------|------------------------------------|-----------|
| <b>1</b> | <b>The basics</b>                  | <b>1</b>  |
| 1.1      | Introduction . . . . .             | 1         |
| 1.1.1    | Motivation and non-goals . . . . . | 1         |
| 1.1.2    | Requirements . . . . .             | 3         |
| 1.1.3    | The very basics . . . . .          | 4         |
| 1.2      | Setting the stage . . . . .        | 6         |
| 1.3      | Distributions . . . . .            | 9         |
| 1.4      | Next-to-leading order . . . . .    | 9         |
| <b>2</b> | <b>DIS in real life</b>            | <b>11</b> |
| 2.1      | DIS is diverse . . . . .           | 11        |
| 2.2      | Interpolation grids . . . . .      | 11        |
| 2.3      | DGLAP . . . . .                    | 11        |
| 2.4      | Scale variations . . . . .         | 11        |
| 2.5      | Heavy Quarks . . . . .             | 12        |
| 2.6      | $\gamma_5$ . . . . .               | 12        |
| 2.7      | TMC . . . . .                      | 12        |
| <b>3</b> | <b>More advanced topics</b>        | <b>13</b> |
| 3.1      | Positivity . . . . .               | 13        |
| 3.2      | Higher orders . . . . .            | 13        |
| 3.3      | QED . . . . .                      | 13        |
| 3.4      | Valo . . . . .                     | 13        |
|          | <b>Bibliography</b>                | <b>15</b> |



# Chapter 1

## The basics

### 1.1 Introduction

#### 1.1.1 Motivation and non-goals

We aim in this course to make a meaningful comparison between the theoretical predictions and the experimental measurement of a fully inclusive Deeply Inelastic Scattering (DIS) cross-section. Instead of an abstract, theoretical discussion about DIS, which can be found in a standard lecture on particle physics, we're looking here to make a connection with real-life physics solving current research questions. The main focus is be naturally on PDF, for which DIS plays a major role. In fact, DIS was, is, and will be a very important constrain in determining Parton Distribution Functions (PDFs) - only recently, e.g. starting from NNPDF4.0 [2], LHC data starts to dominate in PDF fits. Stressing the importance and relevance of PDFs for any kind of perturbative QCD (pQCD) application and to understand the inner structure of hadrons is left to the literature [3–5]. In turn, DIS itself can be used to answer unresolved questions about PDFs, e.g. whether the proton has an intrinsic charm component [6].

In order to obtain a more realistic DIS treatment, we review general pQCD features and exemplify them using DIS. In fact, fully inclusive DIS comes in very handy, since here the effects mostly appear in their simplest form and when considering hadronic collisions, e.g. at the LHC, the strategies are mostly the same just more involved. We keep a phenomenological point of view with a specific focus on the numerical implementation as, eventually, we want to compare to the experimental measurements. We refrain from giving detailed proofs in this course and point to the standard textbooks [7–11] for that.

Instead, we consider different features as given and focus on their practical implications. However, a major complication in this course is the interconnection between almost all features and while we discuss them here in turn, we highlight also how they are intertwined with each other.

In recent years (and much more so hopefully in the coming years), DIS has become a major research topic again thanks to the advent of the Electron Ion Collider (EIC) [12]. While the planned physics program of the EIC goes far beyond standard fully inclusive DIS, this simplest case can serve as an introduction into more complicated topics.

The structure of this course follows in large parts the implementation of Yadism (Yet Another DIS module) [1, 13], which is developed and used by the NNPDF collaboration in their PDF determinations. The online documentation, which is available at

<https://yadism.readthedocs.io/> ,

may serve as additional, more technical introduction into the topic. Note that Yadism is an open-source project to which everybody can contribute<sup>1</sup>.

Since DIS is a very wide field, it is worth spelling out explicitly some non-goals:

- We only consider pQCD in the collinear framework here. However, because DIS is such a powerful experiment, which offers a clean access to the inner structure of the protons it can be studied using other frameworks as well, such as Color Glass Condensates [14, 15] → ask H. Mäntysaari or T. Lappi
- We only consider fully inclusive cross-sections here (to be defined in Section 1.2 and with a tiny exception to be discussed in Section 2.5). In particular, we do not consider any more complicated final states, such as, e.g., single jet or di-jet production [16].
- We only consider pQCD in the collinear framework here. While, Monte Carlo Event generators, such as, e.g., Pythia [17], are rooted in collinear pQCD they eventually go much beyond and we do not consider their details here [18] → ask I. Helenius.
- We only consider DIS here. In particular, we do not consider photo-production or elastic or diffractive cross sections. We define the necessary kinematics for the distinction below in Section 1.1.3.

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<sup>1</sup>e.g. if some documentation can be improved



## Standard Model of Elementary Particles

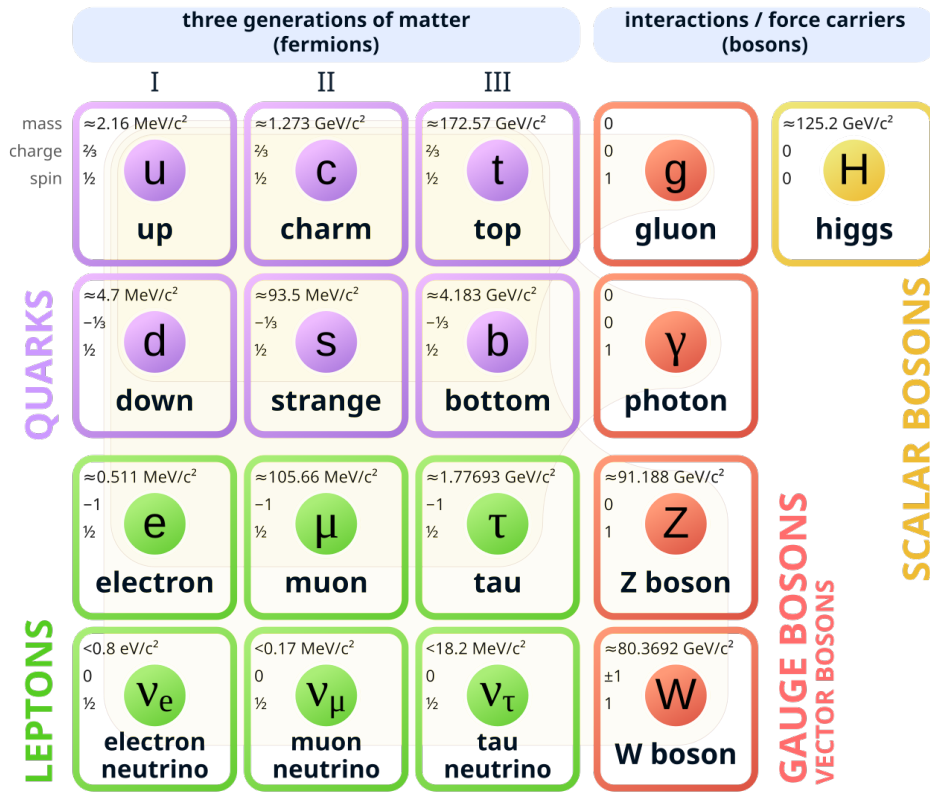


Figure 1.1: The Standard Model of Elementary Particles - taken from [19]

- We compute the theoretical predictions here. So while we try to keep the connection to the experimental side close, we do not discuss any specific details about how to measure cross-sections.

### 1.1.2 Requirements

We need some basic knowledge from particle physics and, in particular, you need to be familiar with the Standard Model overview shown in Fig. 1.1. We discuss in this course each and everyone of these particles, with the exception of the three heaviest in each category: the heaviest quark, the top, the heaviest lepton, the tau, and the heaviest boson, the Higgs. They work, in principle, mostly like their lighter brothers and sisters, but in practice are basically always neglected. We give a more thorough justification to discard them when we get to the actual numbers. All other particles explicitly appear in this course and you need to know their charges - see also Fig. 1.1.

#### Summary forces, particles, and charges

- The carrier of the strong force is the gluon
- Quarks have a strong charge, leptons don't
- The carriers of the electro-weak force are the photon, the Z boson, and the W boson
- Quarks and charged leptons have electro-magnetic charges
- Quarks and leptons have weak charges

We also need some basic knowledge from Quantum Field Theory (QFT).

### 1.1.3 The very basics

Let's start from the top:

#### DIS in words

fully inclusive DIS = fully inclusive deeply inelastic scattering

What does that actually mean? We begin from the back:

- “scattering” means, of course, we are scattering two particles with each other. In this particular case, we are scattering a lepton off a hadron and to make things simple in the beginning we consider an electron is scattered off a proton.
- “inelastic” means we are breaking the target, the proton, apart; it shatters into many pieces.
- “deeply” is an adjective<sup>2</sup> to inelastic and it means we are looking “deep” into the inside of the proton. And in order to resolve small things, as usual, we need an highly energetic probe.
- “fully inclusive” means we are observing a single particle in the final state, which is the electron, but nothing else.

Thus, we can write the same information again with a formula

---

<sup>2</sup>thus it is “deeply inelastic” or “deep-inelastic”, but not a list of adjectives “deep inelastic”

## DIS in a formula

fully inclusive DIS =  $e^-(k) + p(P) \rightarrow e^-(k') + X$  with  $-(k - k')^2 \gg \Lambda_{\text{QCD}}^2$

where we have assigned the four-momenta  $k, P, k'$  to the incoming electron, the incoming proton and the outgoing electron respectively. As usual  $X$  marks the inclusiveness, i.e. we don't observe anything else apart from the electron. By requiring a large momentum-transfer between the incoming and the outgoing electron, i.e. their difference four-momentum vector has a large magnitude, we probe the proton with highly energetic parton (which we can think of as photon).

Next, we need a way to actually compute theory predictions for the cross-sections, which you can measure in an experiment. We use the factorization theorem [20],

## Factorization Theorem

$$\sigma(x, Q^2) = \sum_j \int_x^1 \frac{dz}{z} C_j(z, Q^2) f_j(x/z, Q^2) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right) \quad (1.1)$$

$$= \sum_j (C_j(Q^2) \otimes f_j(Q^2))(x) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right) \quad (1.2)$$

$$= (\mathbf{C}(Q^2) \otimes \mathbf{f}(Q^2))(x) + \mathcal{O}\left(\frac{\Lambda_{\text{QCD}}^2}{Q^2}\right), \quad (1.3)$$

which states how to compute a cross-section  $\sigma(x, Q^2)$  from coefficient function  $\mathbf{C}(z, Q^2)$ , computable in pQCD, and a universal PDF  $\mathbf{f}(\xi, Q^2)$ . In the remaining part of the course we discuss, enhance and correct this formula in many different ways and basically challenge every aspect. We define each and every variable and function in Section 1.2, so we can focus for the moment on two other important aspects: universality and notation.

The crucial, physical aspect about Eq. (1.1) is universality: for any cross-section  $\sigma$  measured at any  $x$  and  $Q^2$  there is exactly one PDF  $\mathbf{f}(\xi, Q^2)$ . Instead, the coefficient functions  $\mathbf{C}(z, Q^2)$  may depend on the definition of the cross-section, but they are computable from first principles. This implies the predictive powers of our framework: if  $\mathbf{C}$  is fixed and  $\mathbf{f}$  is universal, we can measure  $\sigma$  in one set of experiments and then predict the outcome of another set of measurements  $\sigma'$ .

The other, more practical aspect of Eqs. (1.1) to (1.3) is the short-hand notation we introduce there. While Eq. (1.1) is the most canonical form of the factorization theorem, which can be found like this in most textbooks, we introduce in Eq. (1.2) the multiplicative convolution symbol  $\otimes$ . Note that we have suppressed the variable over which the convolution is performed and recall that convolution is associative, i.e.  $(g \otimes h)(x) = (h \otimes g)(x)$ . Finally, in Eq. (1.3) we introduce the vector notation, where we denote object, which have a non-trivial dependence on the parton flavor  $j$ , with bold font. Equation (1.3) looks nice and compact, but it worth recalling that it already hides a lot of details: specifically,  $C_j(z, Q^2)$  is an object which depends on three different variables. We use vector notation and the index notation interchangeably depending on the context.

To Leading Order (LO) accuracy, the coefficient functions  $\mathbf{C}$  greatly simplify and we find

$$\sigma(x, Q^2) = \sum_j e_{q_j}^2 f_j(x, Q^2), \quad (1.4)$$

i.e. the cross-section is given by the sum of the quark PDFs weighted by their electrical charges squared. Note that here the variables, which are used for the experimentally measured cross-section on the left-hand-side, are also used as arguments of the PDFs on the right-hand-side. This is a pure coincidence only valid in the simplest case at LO and two historical comments can be made here. First, when studying older literature this equivalence can lead to confusion as, e.g., PDFs have been also called “structure functions” at the time, or what exactly we mean when talking about a general variable “ $x$ ”. Second, in the so-called “parton model”, which was popular before QCD was invented, the index  $j$  just referred to a “parton” with some properties. Instead, within QCD we now know exactly, what the object in question is: a quark, which has a strict definition in QFT with well-defined properties and quantum numbers. For simplicity, we still call an object, which can be either a quark or a gluon “parton”.

## 1.2 Setting the stage

In the following section we collect and define all ingredients needed for Eq. (1.1). As already said, DIS is given by

$$e^-(k) + p(P) \rightarrow e^-(k') + X \quad (1.5)$$

**FH: copy DIS diagram from PDG** where the four-momenta  $k, P, k'$  belong to the incoming electron, the incoming proton and the outgoing electron respectively. This allows us, first, to define four momentum of the exchanged photon

**FH!**

$$q = k - k' \quad (1.6)$$

and from there the set of four common variables:

$$s = (k + P)^2 = \frac{Q^2}{xy} \quad (1.7)$$

$$Q^2 = -q^2 \quad (1.8)$$

$$x = \frac{Q^2}{2q \cdot P} \quad (1.9)$$

$$y = \frac{q \cdot P}{k \cdot P} \quad (1.10)$$

Note that Eq. (1.7) gives an explicit relation among these, thus we have only three independent and one dependent variable. In practice, the invariant mass of the electron-proton system  $s$  is fixed by the experimental setup, i.e. by the energy of the colliding particles. Also the measurable range of  $y$  is in practice limited by the experimental geometry. DIS cross-sections are typically provided as double-differential cross-sections in  $x$  and  $y$ . Finally, it is convenient to define the mass squared of the system X recoiling against the scatter electron,

$$W^2 = (P + q)^2 = Q^2 \left( \frac{1}{x} - 1 \right) \quad (1.11)$$

#### DIS requirements

We require  $Q^2 \gg \Lambda_{\text{QCD}}^2$  (deep) and  $W^2 \gg \Lambda_{\text{QCD}}^2$  (inelastic) to call a lepton-hadron scattering DIS.

We can decompose the (double-differential) cross-section into a leptonic interaction and a hadronic interaction, yielding each a tensor in Minkowski space,  $L_{\mu\nu}$  and  $W_{\mu\nu}$  respectively. Thus, we can write

$$\frac{d^2\sigma}{dx dy} = \frac{2\pi y \alpha_{\text{em}}^2}{Q^4} L_{\mu\nu} W^{\mu\nu} \quad (1.12)$$

where we have introduced the electro-magnetic coupling  $\alpha_{\text{em}}$ , which appears once on the leptonic side and once on the hadronic side. We can express the leptonic tensor using plain Quantum Electrodynamics (QED) and provide a decomposition of the hadronic tensor in terms of scalar functions. Together, we obtain

$$\frac{d^2\sigma}{dxdy} = \frac{4\pi\alpha_{\text{em}}^2}{xyQ^2} \left( \left(1 - y + \frac{y^2x}{2}\right) F_2(x, Q^2) - \frac{y^2x}{2} F_L(x, Q^2) \right) \quad (1.13)$$

where we have introduced the two common scalar functions, now called structure functions  $F_2$  and  $F_L$ , which are often measured in experiments. It is convenient to also define a third, linearly dependent structure function via

$$2xF_1(x, Q^2) = F_2(x, Q^2) - F_L(x, Q^2). \quad (1.14)$$

Keep in mind that the discussion so far is a major simplification of DIS and in Section 2.1 we reintroduce the main complexity.

Let us continue to collect the remaining ingredient for Eq. (1.1):

- The sum over partons  $j$  includes quarks and gluons and we refine in Section 2.5 how large the sum actually is.
- The parton momentum fraction  $\xi = x/z$  is the fraction of the four momentum the parton carries with respect to its parent proton.
- The probability of such a configuration to occur is given by the PDF, i.e.  $f_j(\xi, Q^2)d\xi$  is the probability to find a parton  $j$  with momentum  $\xi P$  inside a proton with momentum  $P$  at an experiment performed at the scale  $Q^2$ .
- The integration over  $z$  sums all configuration to generate the final state configuration.

Finally, the coefficient functions  $\mathbf{C}(x, Q^2)$  are computable order-by-order in pQCD and thus can be decomposed as

$$\mathbf{C}(x, Q^2) = \sum_{k=0} \alpha_s^k(Q^2) \mathbf{C}^{(k)}(x). \quad (1.15)$$

Note that the  $Q^2$  dependency of  $\mathbf{C}(x, Q^2)$  is fully captured by the strong coupling  $\alpha_s(Q^2)$ .

### 1.3 Distributions

As we have seen the coefficient functions may contain distributions. For example the quark coefficient function at LO is just a Dirac delta distribution:

$$C_q^{(0)}(z) = \delta(1 - z) . \quad (1.16)$$

To better compare with our next step, let's repeat the definition of the Dirac delta distribution:

$$\int_0^1 dz \delta(1 - z) f(z) = f(1) . \quad (1.17)$$

However, this is only the simplest case, which appears at LO, and at higher order we get more complicated distributions. For example the quark coefficient function at NLO is given by

$$C_q^{(1)}(z) = C_F \left[ 4 \left( \frac{\ln(1 - z)}{1 - z} \right)_+ - 3 \left( \frac{1}{1 - z} \right)_+ - (9 + 4\zeta_2) \delta(1 - z) - 2(1 + z) \ln \left( \frac{1 - z}{z} \right) - 4 \frac{\ln(z)}{1 - z} + 6 + 4z \right] \quad (1.18)$$

where the first line only contains distributions. Here, we have introduced the usual  $+$ -distributions, which are defined via

$$\int_0^1 dz a(z)_+ f(z) = \int_0^1 dz a(z) (f(z) - f(1)) . \quad (1.19)$$

We review in Section 1.4 again briefly on how these distributions appear, but for the moment we turn to more practical matters as the goal of our course is to get an actual comparison to experimental data. The problem is: distributions are (complicate) math objects, defined only via integral relations, Eqs. (1.17) and (1.19), which are unsuitable for numeric evaluation.

### 1.4 Next-to-leading order

- NLO<sub>q</sub>

- NLOg

We need the Riemann Zeta function  $\zeta(n)$  below, which is defined at integer values  $n$  by

$$\zeta_n = \zeta(n) = \sum_{j=1}^{\infty} \frac{1}{j^n} \tag{1.20}$$



# Chapter 2

## DIS in real life

### 2.1 DIS is diverse

- $\gamma/Z$
- leptons
- hadrons: p+d+A; nucl. PDF
- experimental overview

### 2.2 Interpolation grids

### 2.3 DGLAP

- resummation
- Mellin

### 2.4 Scale variations

- scheme A/B/C
- MHOU

- covariance matrix

## 2.5 Heavy Quarks

- $F_2^c$
- FONLL

## 2.6 $\gamma_5$

- helicity
- HVBM + Larin

## 2.7 TMC

# Chapter 3

## More advanced topics

### 3.1 Positivity

- DIS vs. MSbar

### 3.2 Higher orders

- IHOU
- N3LO
- channels
- loops and legs

### 3.3 QED

- Qiu et al.

### 3.4 Valo

- $C_\gamma \rightarrow k$



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